PHAS0050 : Climate & Energy
Module 2 : Energy Generation

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1 Introduction

In this part of the course we consider briefly the motivation of the search for carbon neutral and/or renewable energy sources, discuss in detail supply options, investigate demand reduction and attempt to match supply and demand between now and the year 2050. We shall be concentrating on these issues from the UK perspective. These notes have been heavily influenced by the books of MacKay [1] and Andrews & Jelley [2] but we do deal with certain topics differently and often go beyond these books.

The motivations driving the search for carbon neutral and/or renewable energy sources are threefold :

- fossil fuels are finite resources although new exploitation techniques continue to extend the depletion dates;
- there should be security of energy supply;
- the probability is high that the burning of fossil fuel will lead to climate change. Doing nothing is too high a risk to take.

Planning a global energy strategy was above all bedevilled by the fact that 55% of the CO_2 is emitted by the USA and China, nations that did not sign up to the Kyoto protocols (December 1997). At the Lima Conference (Dec. 2014) there were important announcements from China and the US following a meeting of the heads of state, Presidents Xi and Obama. China forsees that its CO_2 emission will peak in 2030 and that its current program of building 'one new coal fired plant every week' will be cut back so that its current electricity generation by coal burn be replaced by nuclear ($60\ \mathrm{GW_e}$) and renewables ($740\ \mathrm{GW_e}$). The Chinese are serious players in nuclear reactor construction and in pursuing alternative designs such as molten salt reactors. The decarbonisation program of electricty generation is being driven by the need to reduce serious air polution and the realization that the temperature rise in China (greater than the global average) has serious consequences for agriculture and for coastal communities.

In the US President Obama has taken executive action to put new restrictions on carbon emissions from power stations so as to eliminate the coal burn by 2025. One can be more confident of Chinese actions bearing in mind the democratic process in the US, hamstrung as it is by political impotence, vested interests and a strong constituency of climate change sceptics.

Canada also withdrew from the Kyoto protocol since it wishes to protect its exploitation of the Athabasca shales to produce oil (see later) and have sold the state backed company Atomic Energy of Canada Ltd. (AECL) which was responsible for the successful CANDU reactors for a derisory price. Australia's government has also recently been 'climate change sceptic' - it has huge coal reserves which it would like to export.

Fracking both for gas (which produces less than half the CO_2 emission of coal combustion) and oil has had a huge effect on the global politico - economic balance of power. The US will be largely self sufficient in oil and gas which threatens process plant capital investment in Europe where the gas price is two to three times that in the US. Saudi Arabia seems intent on bankrupting the US oil shale industry by forcing the oil price below $\$70$ a barrel, the break even price for shale oil, an action made easier by the oil glut caused by the economic slow down in China. The falling oil price seriously undermines the Russian economy but does not yet weakens Russia's position in the conflict over the Ukraine with the West. Indeed fossil fuel resources seem to be increasing all the time and it is clear that one third to one half of the known fossil resources must remain underground to avoid a global $2°\mathrm{C}$ temperature rise, with colossal implications for producer states and international energy companies alike. Technologies that might greatly reduce the CO_2 emission by Carbon Capture and Sequestration has just started on an industrial scale at a $110\mathrm{MW_e}$ coal fired unit in Canada and is expected to carry an efficiency penalty of 20%.

Another important development is that electricity generated by Photo Voltaic (PV) solar panels is now 'grid competitive' for countries located at low latitudes. Should large scale, cheap storage methods become

available then such countries would have a viable electricity supply. All other electricity supply from renewable or carbon neutral sources are more expensive than the current still largely fossil supply. It should also be remembered that a glut of supply from even these more expensive sources leads to a fall in the market price of the generated electricity making even conventional generation uneconomic.

Since the circumstances of individual nations are so very different strategic planning must be at the national level but ideally within an international framework. The UK has a strategy, albeit with considererable uncertainties, to reduce CO_2 emission by 80% by 2050. In 2013 the UK suceeded in getting international construction consortia and the funding for the build of two new EPR reactors at Hinckley Point led by EDF, albeit at a high price once the electricity is supplied. However doubts are creeping in - the contractor is the French company AREVA who is constructing a new reactor at Flammanville for EDF and at Olkiluoto for the Finnish authorities. Both projects are five to nine years late and far over budget. In addition, serious deficiences have been discovered in the steel of parts of the pressure vessel of the Flammanville reactor (April 2015). China is a 40% investor in these projects and was already becoming restive given that delays are also being experienced in the two EPR reactors being constructed in China by the French consortium. The latest revelation will no doubt increase the pressure to include Chinese component parts and could lead to an iternational takeover of AREVA.

Following the Fukushima nuclear accident Germany decided to run down its nuclear generation with the result that it increased its coal burn by buying cheap coal whose price had fallen due to the fracking boom in the US. Germany has also made substantial investments in off shore wind generation in the north and in solar energy in the south, with the result that sudden surge generation from the north or from the south causes the market price of electricity to drop, making conventional generation often uneconomic.

Action on climate change in recent years has been influenced by the international debt crisis, political impotence, the Fukushima nuclear accident, large scale fracking in the US and the realisation that nuclear and renewable energy with the possible exception of PV solar generation, are expensive. To be blunt, under the Kyoto protocols there has been little global progress since 1997 - in fact the CO_2 concentration of $370\ \mathrm{ppm}$ in 2000 has increased to $397\ \mathrm{ppm}$ in 2015. The Paris Agreement, which commits nations to "[hold] the increase in the global average temperature to well below $2°\mathrm{C}$ above pre-industrial levels", came into force. In principle this is a major step forward, although criticisms have been raised regarding the lack of binding enforcement mechanisms.

From Figure 1 showing the Carbon Dioxide (CO_2) concentratation by volume (in parts per million) over the past 1100 years measured from air trapped in ice cores up to 1977 and directly in Hawaii (from 1958) onwards, something new may have indeed happened between 1800 AD and 2000 AD. In 1769 James Watt patented his steam engine which was much more efficient than the engines invented some 70 years earlier. Fossil fuels then powered the Industrial Revolution together with the concomitant economic growth. The so called 'hockey stick' temperature rise was called into question by climate change sceptics who obtained e-mails hacked from the University of East Anglia Climate Research Unit. However a group of such sceptics, the Berkeley Earth Temperature team have recently completely confirmed the 'hockey stick' behaviour. In 2000 26 gigatonnes of CO_2 (or 34 gigatonnes $(CO_2)_{\mathrm{eq}}$ of greenhouse gases including CO_2, methane, and nitrous oxides (the latter two expressed in equivalent amounts of $(CO_2)_{\mathrm{eq}}$) were sent per year into the atmosphere largely by the burning of fossil fuels. The biosphere and the oceans send far larger amounts, 330 and 440 gigatonnes per year respectively but these flows are balanced by the CO_2 flows back from the atmosphere. Please note we do not differentiate between 'tonnes' and 'ton' as the difference is small and the year 2000 is chosen as representative and current, since the statistics were last surveyed and completed for that year. How does this flow rate compare with the CO_2 currently in the atmosphere ? We first calculate the total mass of air in the atmosphere M_{atmos} taking the atmospheric pressure p_{atmos} to be 10^5 Pa, i.e. $10^5\ \mathrm{N\,m^{-2}}$, the Earth's radius R_E to be $6378\ \mathrm{km}$ and the acceleration due to gravity, $g = 9.81\ \mathrm{m\,s^{-2}}$. Then :

$$M_{\mathrm{atmos}} g = 4\pi R_E^2 \times p \quad \text{which yields} \quad M_{atmos} \simeq 5 \times 10^6 \text{ gigatonnes.}$$

Therefore, if the current concentration of CO_2 is $390\ \mathrm{ppm}$ by volume or $582\ \mathrm{ppm}$ by mass (the mass

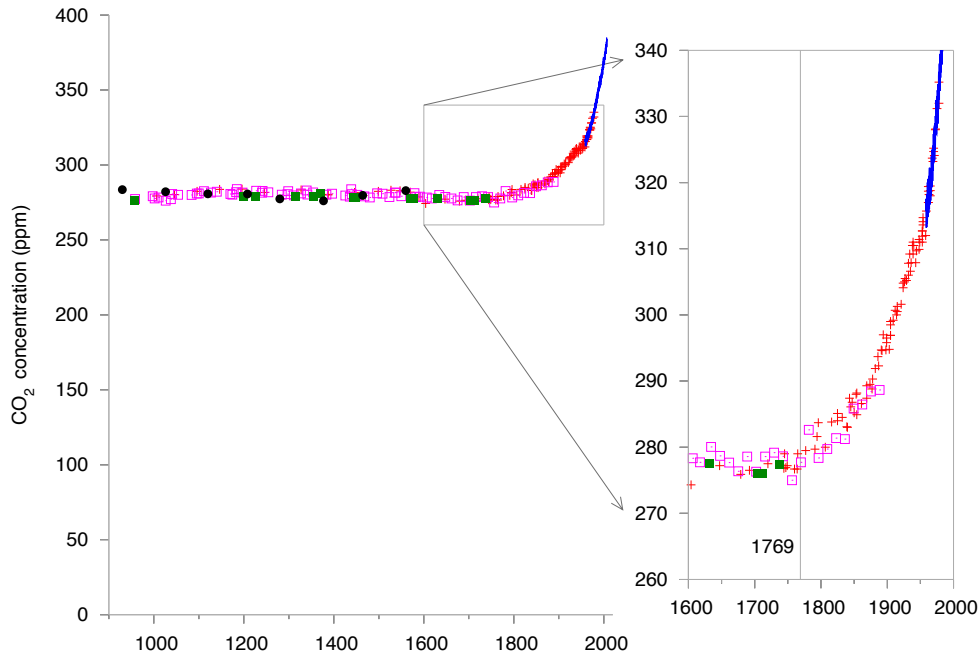


Figure 1: CO_2 Concentration vs. year

concentration is related to the volume concentration by the ratio of the molecular masses 44 for CO_2 and 29 for the mean molecular mass of air) then there is close to 3000 gigatonnes of CO_2 in the atmosphere. The Earth's population in 2000 was 6 billion (it's now 7 billion), or 6×10^9, thus 34 gigatonnes of $(CO_2)_{\rm eq}$ yearly emissions correspond to an emission rate of $5\frac{1}{2}$ t yr^{-1}. However as Mackay's book says all people are created equal but we dont all emit $5\frac{1}{2}$ tonnes of CO_2 per year - region by region and country by country the yearly emission per person is very different as shown in Figures 2 and 3. In the USA the CO_2 equivalent emmission was close to 24 tonnes, in the UK 11 tonnes, whereas in China it was close to 4 and in India 2 tonnes per year per person (note : the global recession, fracking in the US, and the export of manufacture to China from the US and Europe, have significantly changed these numbers to 17.3 tonnes in the US, 7.5 tonnes in Europe and 7.2 tonnes in China). If we expect the ethical idea that the 'polluter should pay'

China	1.3 Billion
India	1.1 Billion
USA	300 million
Japan	127million
UK	60 million

Table 1: Some population numbers in the year 2000.

then we should ask how big is each country's historical footprint. Figure 4 shows each country's cumulative emissions of CO_2 tonnes per year per person expressed as an average rate over the period 1880-2004. It is clear that the lion's share of the cumulative pollution is caused by the industialised nations. To have a reasonable chance of avoiding a global temperature rise of more than 2°C then CO_2 emissions must fall by 5% per year from 2000 until it is $\simeq 1$ tonne per year per person by the year 2050. So in the UK more than a 90% reduction from 11 tonnes per year per person and in the USA about a 95% reduction from 24 tonnes per year per person from 2000 levels are required in CO_2 emission. However it is now clear that this target

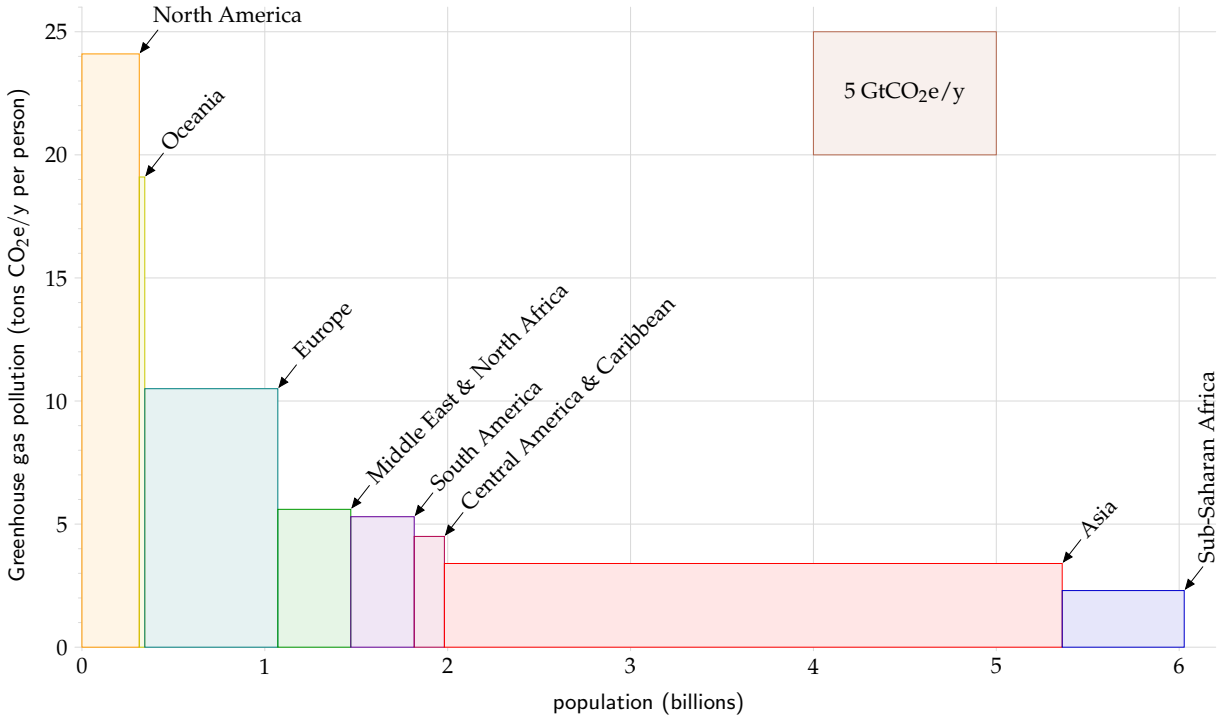


Figure 2: CO_2 Greenhouse gas pollution vs. regional population

is going to be missed. Nevertheless the message is clear - over the next 30 years energy production by fossil fuels must either be reduced drastically to be 'Carbon Neutral', or perhaps involving 'Carbon Capture'.

2 A Word about Units

In this course it is particularly important to follow the philosophy of MacKay's book and to choose units such that a physically relevant quantity turns out to be a small number of such units. For example it is easier to remember and to understand that the current rate of $(CO_2)_{eq}$ emissions is $5\frac{1}{2}$ t yr^{-1} rather than the global rate of 34 gigatonnes $(CO_2)_{eq}$ yearly.

For **energy** we use the kilowatt-hour (kWh) which is the energy unit on domestic electricity and gas bills. Centrica quotes 14p per $\mathrm{kWh_e}$ of electricity and a standing charge of 16p per day. On gas bills the corresponding costs are 5.5p per kWh and a 26p standing charge. Note that the current (2014) *wholesale* prices in the UK are $5.5\ \mathrm{p}\ (\mathrm{kWh_e})^{-1}$ for electricity and $2\mathrm{p}\ (\mathrm{kWh})^{-1}$ for gas, both reflecting historic supply and generating capacity. Gas is a primary fuel whereas most electricity is derived from a primary fuel by means of heat engines which are limited by thermodynamic inefficiency (see later). Also electricity generation and distribution involve higher capital costs. Recall that one watt (1 W) is the power of one Joule per second. So a kilowatt hour is $1000 \times 60 \times 60$, i.e. 3.6 MJ.

For **power** we use the kilowatt-hour per day ($\mathrm{kWh\ d^{-1}}$) unit. The light and heat generated by a 40 W incandescent bulb (you can't buy these any longer!) has a power close to one $\mathrm{kWh\ d^{-1}}$. The energy per day (i.e. the power in $\mathrm{kWh\ d^{-1}}$) produced by a 40 W bulb is $40 \times 60 \times 60 \times 24$ J $\mathrm{d^{-1}}$ i.e. 3.456×10^6 J, i.e. $\simeq 1\,\mathrm{kWh\,d^{-1}}$ (see the end of the previous paragraph). As Mackay also points out a $\mathrm{kWh\ d^{-1}}$ is the energy per day or power you would get out of a servant assuming you had one! You could also work out the efficiency of your servant given that he or she would need a daily food intake of at least 2500 kcal where

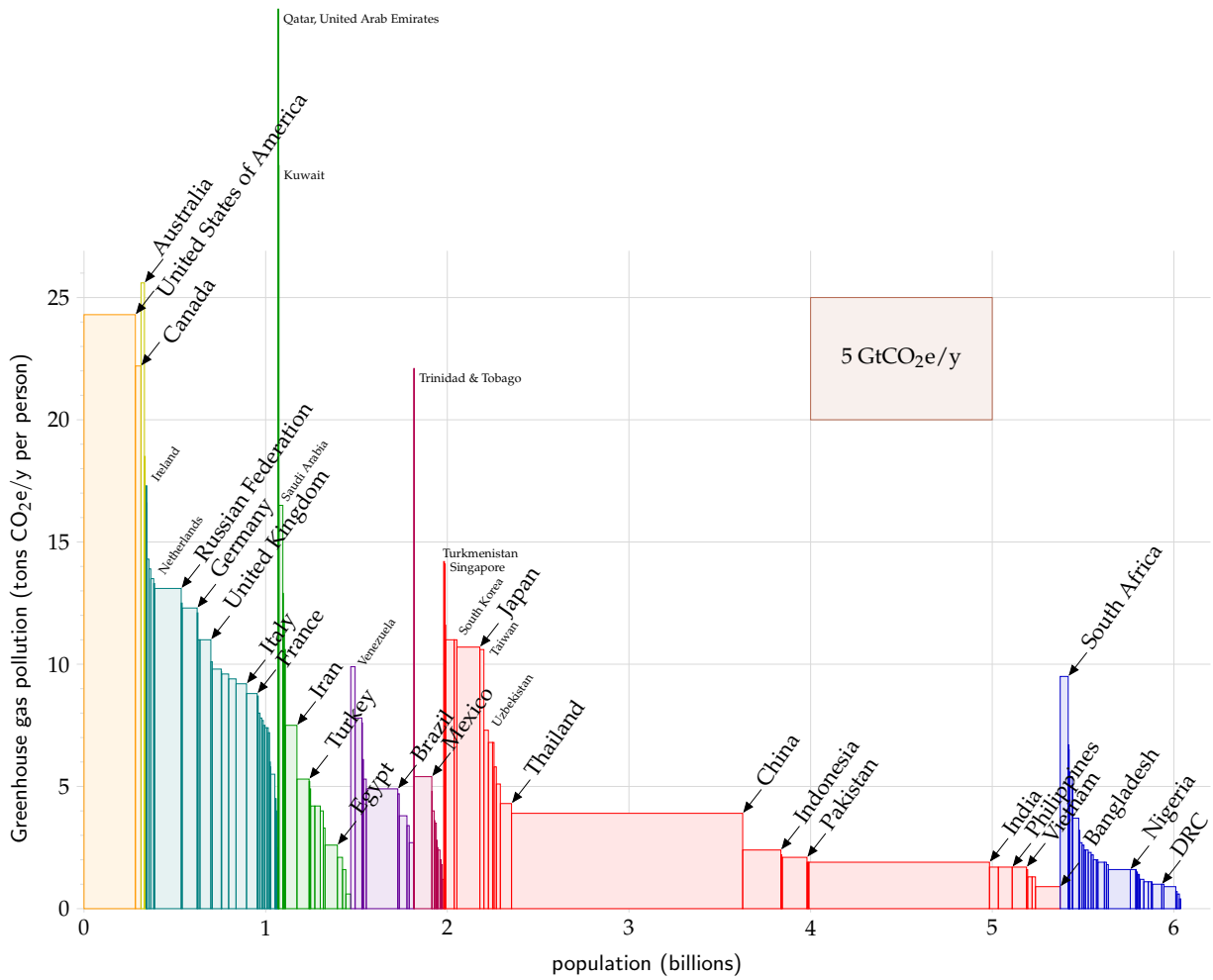


Figure 3: CO_2 Greenhouse gas pollution vs. national population

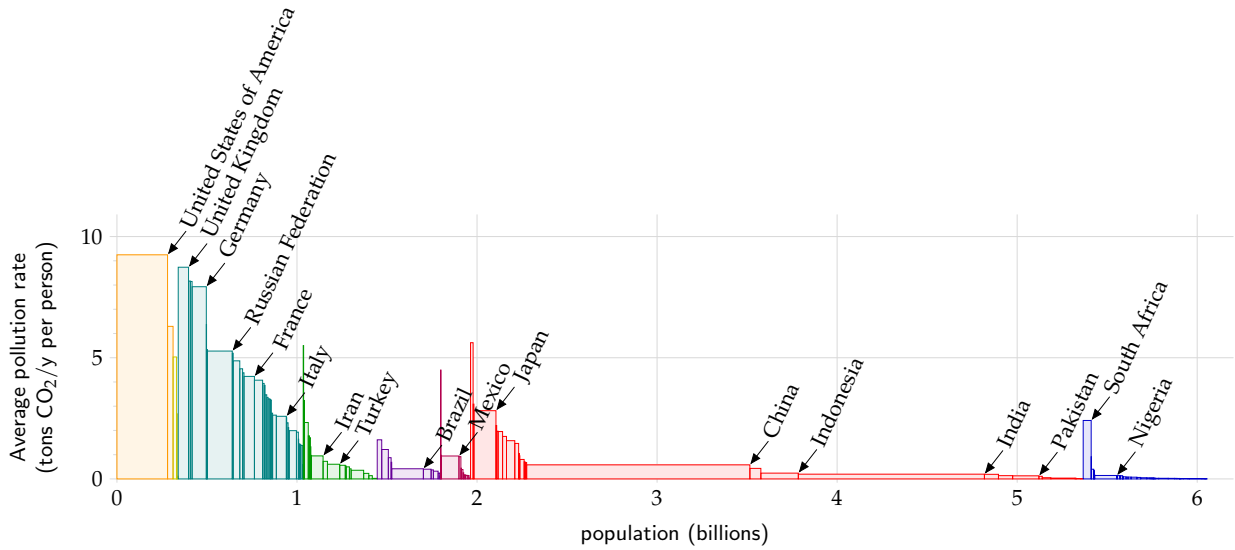


Figure 4: CO_2 Cumulative greenhouse gas pollution vs. national population

$1\ \mathrm{cal} = 4.18$ J i.e. the daily food energy intake is $10.4\ \mathrm{MJ}$ or $2.9\ \mathrm{kWh\ d^{-1}}$ per day, yielding an efficiency of about 34%! We could also readily argue that the power of one $\mathrm{kWh\ d^{-1}}$ is an energy of one kWh in 24hrs. so the power is one kWh /24h. or 1000 W/24, i.e. close to the 40 W we have previously derived.

In the atomic and nuclear world the natural energy unit is the electron volt eV which is the energy gained by an electron of charge 1.6×10^{-19}C falling through an electric potential of 1 volt ie 1.6×10^{-19}J so $1\ \mathrm{eV} = 1.6 \times 10^{-19}$ J.

3 UK and Global Primary Energy Consumption.

By 'primary energy' we mean energy released by fossil and nuclear fuels, hydro electricity etc. and does not include energy embedded in foodstuffs nor energy embedded in imported manufactures. The power embedded in imports (and the associated CO_2) emissions is estimated to be $40\ \mathrm{kWh\ d^{-1}}$ per person in the UK and would appear in the primary power consumption of the exporting nation, which raises interesting political and ethical issues.

Figure 5 shows the national primary power consumption in kWh/d per person (i.e. $\mathrm{kWh\ d^{-1}\ p^{-1}}$) versus annual per capita GDP. We learn from the plot that the power consumption per day per person in the UK is $125\ \mathrm{kWh\,d^{-1}\ p^{-1}}$, $250\ \mathrm{kWh\ d^{-1}p^{-1}}$ in the USA and about $40\ \mathrm{kWh\ d^{-1}p^{-1}}$ in China, and note the correlation between GDP and energy consumption. Note also that the world average power consumption is $60\ \mathrm{kWh\ d^{-1}\ p^{-1}}$. Various sector plots are also presented showing how the primary and final energy consumption is divided by source, sector and end use in the UK. In 2001 the primary energy usage was 18% to services, 25% to industry, 26% to transport and 31% domestic. The fuels consumed were 1% renewables and waste, 9% mainly nuclear electricity, 40% natural gas and 17% coal. Since then the fraction generated by natural gas for example has greatly increased and it is obvious that large CO_2 emission savings are to be sought in the space heating and transport sectors.

MacKay summarises the per person primary power consumption of the UK in 2007 as heating ($40\ \mathrm{kWh\ d^{-1}\ p^{-1}}$), electricity consumption ($18\ \mathrm{kWh\ d^{-1}\ p^{-1}}$, which requires a $45\ \mathrm{kWh\ d^{-1}\ p^{-1}}$ primary power consumption) and transport ($40\ \mathrm{kWh\ d^{-1}\ p^{-1}}$), which adds up $125\ \mathrm{kWh\ d^{-1}\ p^{-1}}$ primary power per person. Note that a primary power consumption of $125\ \mathrm{kWh\ d^{-1}\ p^{-1}}$ corresponds to a power of $125\ \mathrm{kWh\ d^{-1}\ p^{-1}/24\ h}$, i.e. close to $5\ \mathrm{kW\,p^{-1}}$. This simplification ignores some sizeable details such as agriculture and industry and is in part the reason why MacKay adds $12\ \mathrm{kWh\ d^{-1}\ p^{-1}}$ to the primary energy per day per person that he considers achievable in Europe by the year 2050 (see discussion below).

We should also consider how the requirements of CO_2 emission reduction and the energy supply target which MacKay takes to be that the global per capita primary energy consumption converges to about $80\ \mathrm{kWh\ d^{-1}\ p^{-1}}$, can be achieved while allowing for the essential rise in the GDP of developing nations. MacKay claims that it should be possible for the UK and Europe to achieve a carbon free or neutral primary power consumption of $68\ \mathrm{kWh\ d^{-1}\ p^{-1}}$ and he then adopts a higher target of $80\ \mathrm{kWh\ d^{-1}\ p^{-1}}$ to allow for industrial and agricultural activities.

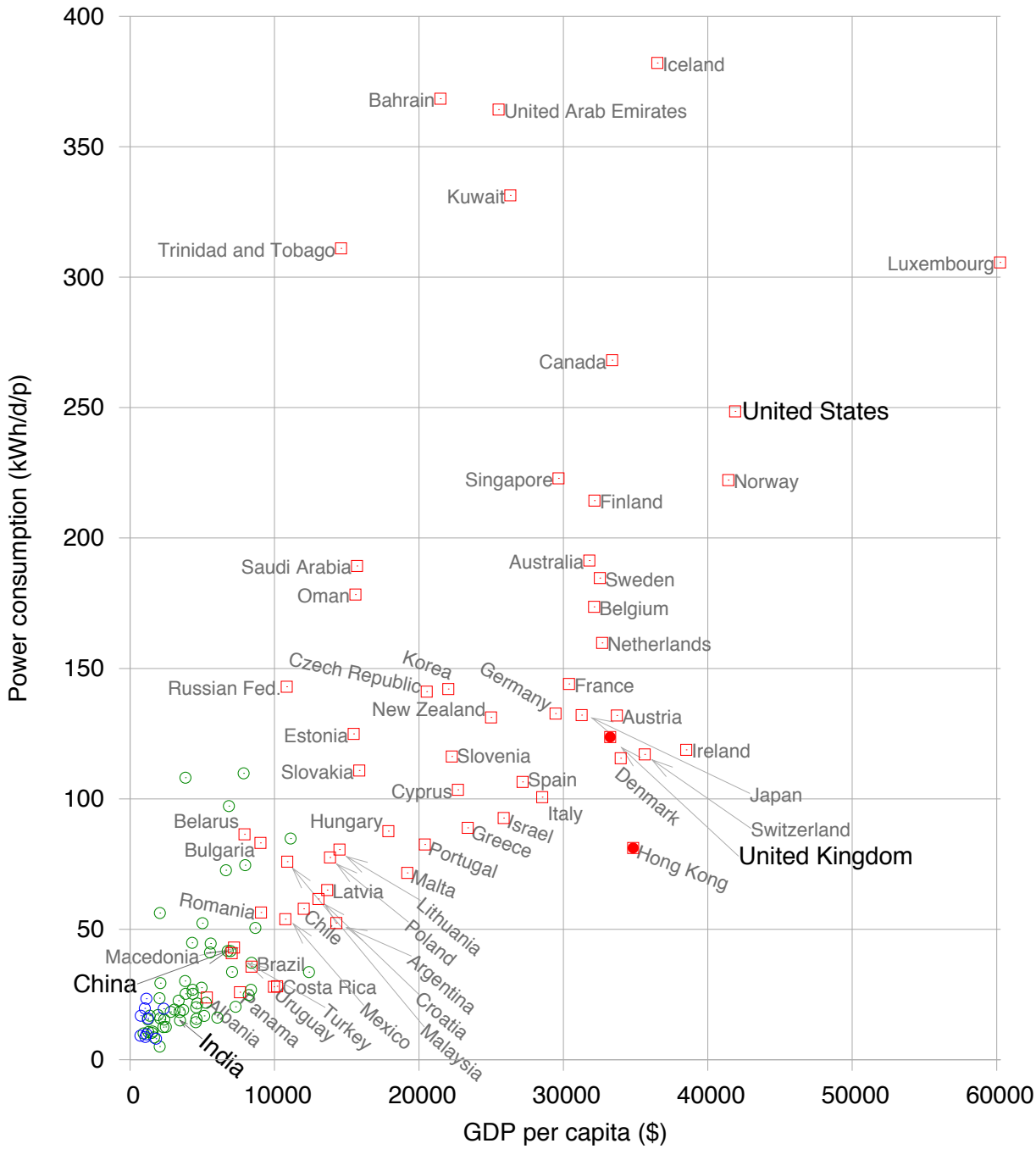
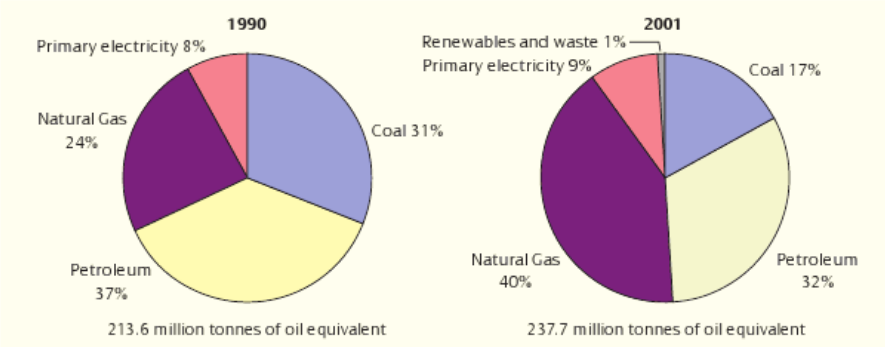


Figure 5: CO_2 Power consumption (kWh/d/p) v. GDP per capita per year (US dollars)

Chart 1.2
Primary energy consumption by fuel, 1990 and 2001

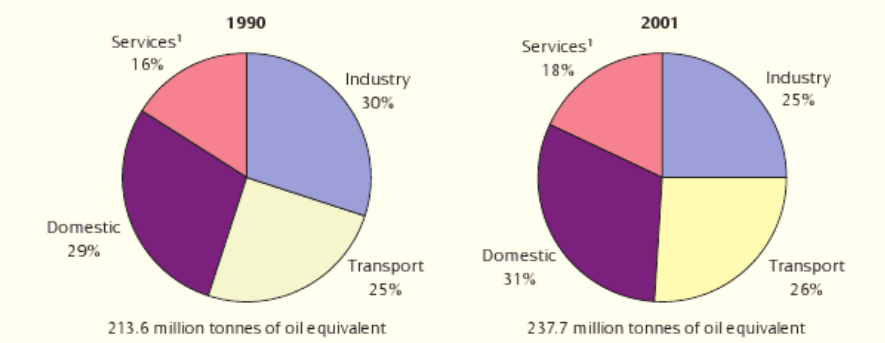


Source: DTI

- Natural gas contributed towards 40 per cent of total primary energy consumption in 2001.
- Petroleum contributed towards 32 per cent of total primary energy consumption in 2001.

1.3 The total amounts of energy consumed by sector in 1990 and 2001 (in primary energy equivalents) are shown in Chart 1.3. Industrial energy consumption fell by 5 per cent between 1990 and 2001 while energy consumption in the transport, domestic and service sectors increased by 18 per cent, 17 per cent and 19 per cent respectively. In primary energy equivalents in 1990 industry was the largest sub-sector, followed by the domestic sector. A decade later the domestic sector was the largest, with transport second. Paragraph 1.6 provides more information about the contribution that each sector makes in final energy terms.

Chart 1.3
Final energy consumption, by sector, in primary energy equivalents, 1990 and 2001



Source: DTI

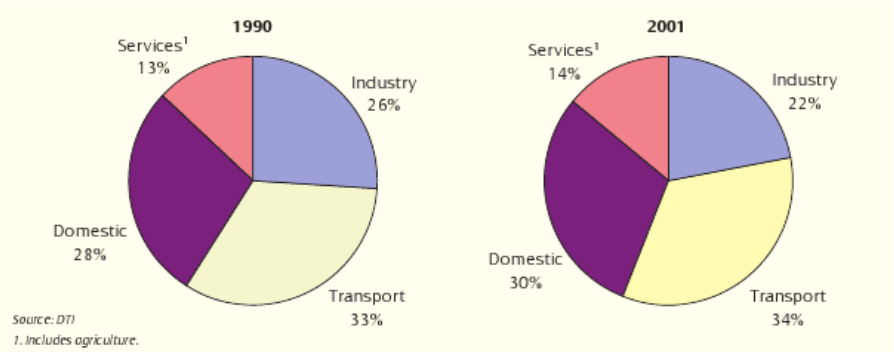
1. Includes agriculture.

- The domestic sector contributed towards 31 per cent of final energy consumption, in primary energy equivalents in 2001.
- The transport sector contributed towards 26 per cent of final energy consumption, in primary energy equivalents in 2001.

Figure 6: CO_2 Sector plots of consumption.

there have been more major changes since 1970, reflecting the shift from energy-intensive industry to the service sector and growth in the transport sector.

Chart 1.5
Percentage sector shares in total energy consumption, 1990 and 2001

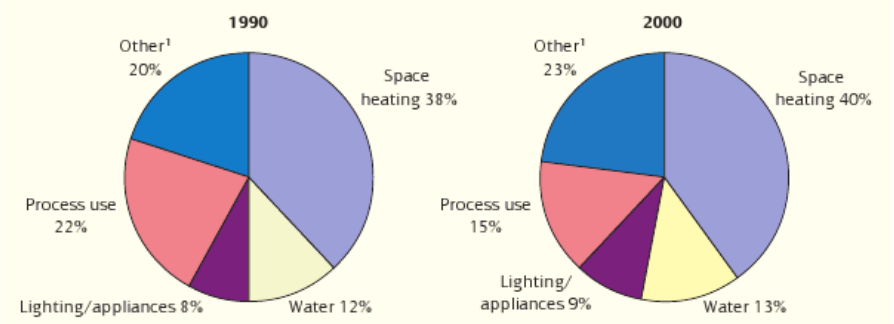


Source: DTI
1. Includes agriculture.

- The transport sector contributed towards 34 per cent of overall energy in 2001.
- The domestic sector contributed towards 30 per cent of overall energy in 2001.

1.7 The amount of energy consumed by each sector can be analysed by how it is used. In 2000, 40 per cent was used for space heating, and 15 per cent for process use. Space heating and hot water accounted for 82 per cent of domestic use of energy and 64 per cent of commercial use of energy in 2000. Chart 1.6 shows how energy consumption by end use changed between 1990 and 2000.

Chart 1.6
Final energy consumption by end use, 1990 and 2000



Source: DTI estimates derived from data provided by FES and BRE
1. Includes cooking, catering, motors, drivers, drying and separation.

- Space heating contributed 40 per cent of all non-transport energy consumption in 2000.
- Process use contributed 15 per cent of all non-transport energy consumption in 2000.

Figure 7: CO_2 Sector plots of consumption.

4 Fossil Fuel Combustion

4.1 Introduction

Fossil fuels have powered the Industrial Revolution and even now most space heating, transport and electricity generation are powered by the combustion of oil, gas or coal. We first review some of the basic combustion processes and some 'Heat and Thermodynamics' which make plain that the production of CO_2 and waste heat are unavoidable. The technology and the sustainability of the resources, how and at what cost can the CO_2 be captured are then discussed. The history of energy technology is well described in the introduction of [2].

4.2 Examples of Fossil Fuel Combustion Reactions

The chemical/combustion reactions :

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O(l) + 55 \text{ MJ per kg. methane, and :}$$

$$C + O_2 \rightarrow CO_2 + 32 \text{ MJ per kg. carbon,}$$

describe the combustion of methane and carbon and give the energy released per kilogram of the fuel at 25°C assuming that water returns to the liquid state $(H_2O(l))$. Calorimetric measurements determine these reaction energies which reflect the change in the chemical bond energies. For completeness we include the corresponding reaction for the combustion of octane, a typical component of crude oil :

$$2C_8H_{18}(l) + 25O_2(g) \rightarrow 16CO_2(g) + 18H_2O(g) + 44.4 \text{ MJ per kg. octane.}$$

From the first reaction we can conclude that the burning of 1 kg of methane produces $44/16$, i.e. 2.75 kg of CO_2 and $55/3.6$ kWh, i.e. 15.3 kWh so that the production of CO_2 is 0.18 kg per kWh. For carbon we find that the combustion of 1 kg of carbon produces 3.7 kg of CO_2 and 9 kWh and that the production of 1 kWh of primary energy by the complete burning of carbon produces 0.4 kg of CO_2. We find also that the combustion of 1 kg of octane produces $8 \times 44/114$, i.e. 3.09 kg of CO_2 and $44.4/3.6$ kWh, i.e. 12.33 kWh so that the production of CO_2 is 0.25 kg per kWh. Note also that a 'barrel of oil' is 136 kg so the energy content of a barrel of oil is about 1700 kWh.

The finest Welsh coal (no longer deep mined) is anthracite and has a carbon content of over 90% whereas Lignite (a dirty fuel, much of it being burnt in Germany following Fukushima) has roughly half this amount.

The value given for methane is the 'Higher Heating Value' (HHV) since it is assumed that after combustion the steam is condensed back to water releasing latent heat. The 'Lower Heating Value' (LHV) assumes that the steam is exhausted after combustion and this is why condensing boilers are more efficient. For hydrocarbons the difference between HHV and LHV values is about 10% and is dominated by the contribution of the latent heat of water. Overall, it is clear that the combustion of methane produces about half the CO_2 emission per kWh primary energy than carbon.

For completness we include here the important non fossil combustion of hydrogen :

$$H_2 + 0.5O_2 \rightarrow H_2O + 143 \text{ MJ per kg. hydrogen } (39.7 \text{ kWh kg}^{-1}; 286 \text{ kJ mole}^{-1}).$$

5 Thermodynamic Cycles

5.1 Basic Heat and Thermodynamics

The fundamental concepts of Heat and Thermodynamics underpin much of this course's content. You should revise and even extend your knowledge of these concepts in this part of the notes.

Recall that the quantity of heat ΔQ required to raise the temperature of a substance by ΔT is given by $\Delta Q = c\Delta T$ where c is the specific heat (or specific heat capacity or heat capacity), i.e.the specific heat is the heat Q required to raise the temperature T of a unit mass of the substance through one degree Kelvin. The molar specific heat refers to the heat required to raise one mole of the substance through one degree K. Except at low absolute temperatures the molar specific heats of solids and liquids are largely constant and reflect the three translational and the internal degrees of freedom of molecules. At the molecular level the mean energy of each degree of freedom is $\frac{1}{2}kT$ (therefore $\frac{1}{2}RT$ per mole) where k is the Boltzmann constant (and R the gas constant), and T the absolute temperature. For water the specific heat is 4.187 kJ kg^{-1} K^{-1} which is higher than any other common substance. The specific heat of gases again reflect the excited internal degrees of freedom but account must be taken whether the heating occurs at constant volume or constant pressure (see below).

During a change of phase, heat is absorbed and the temperature remains constant. The heat ΔQ required to change the phase of unit mass of the substance is called the specific latent heat l and is given by $\Delta Q = l$. As for specific heat, the latent heat can be expressed per mole, the units making it clear as to what quantity of the substance is involved and with upper case symbols used. For water the latent heat of vapourization at $100°$C is 2260 kJ kg^{-1} and the latent heat of fusion at $0°$C is 334 kJ kg^{-1}.

The first law of thermodynamics states that the heat Q *into* a system equals the change in the internal energy ΔU plus the work done W *by* the system, i.e. :

$$Q = \Delta U + W$$

which is a statement of energy conservation in thermodynamic systems. Thermodynamics starts by considering reversible processes where systems are always close to thermodynamic equilibrium and the simplest system is that of the ideal gas (see below) contained in a cylinder with a piston at one end. Work is done by the ideal gas whose pressure p exerts a force $p \times A$ on a piston of cross section A as the piston moves outwards. If the piston moves a distance dx then the element of work done by the system is $pAdx$ or pdV where dV is the change in the volume V of the system. For differential changes in such a system we have :

$$dQ = dU + pdV \ .$$

Note that dQ depends on the path of the thermodynamic process and that the above relationship is only valid for reversible processes where the infinitesimal work done by the piston can be reversed such that the system and its surroundings can return to the initial state. If the internal energy is that of a unit mass of gas then we use the lower case symbol u which is called the specific internal energy.

An ideal gas has an internal energy U dependent only on the the temperature T and obeys the equation of state $pV = nRT$, where n is the number of moles and R is the gas constant ($R = 8.31$ J K^{-1} mol^{-1}). For the molar specific heat at constant volume use $dQ = dU + pdV$, set $dV = 0$ and then it follows that $C_V = (\partial \mathrm{Q}/\partial \mathrm{T})_\mathrm{V} = \partial \mathrm{U}/\partial \mathrm{T}$. It also follows that if a unit mass of ideal gas is heated from T_1 to T_2 *at constant volume* then the change of internal energy is given by $\Delta u = c_V(T_2 - T_1)$.

For the molar specific heat at constant pressure we start from the equation of state $pV = RT$ and differentiate to give :

$$pdV + Vdp = RdT \text{ , such that } pdV = RdT \text{ at constant pressure.}$$

From the first law of thermodynamics we have :

$$dQ = dU + pdV = dU + RdT \text{ at constant pressure and } C_p = \left(\frac{\partial Q}{\partial T}\right)_p = \frac{\partial U}{\partial T} + R = C_V + R \ .$$

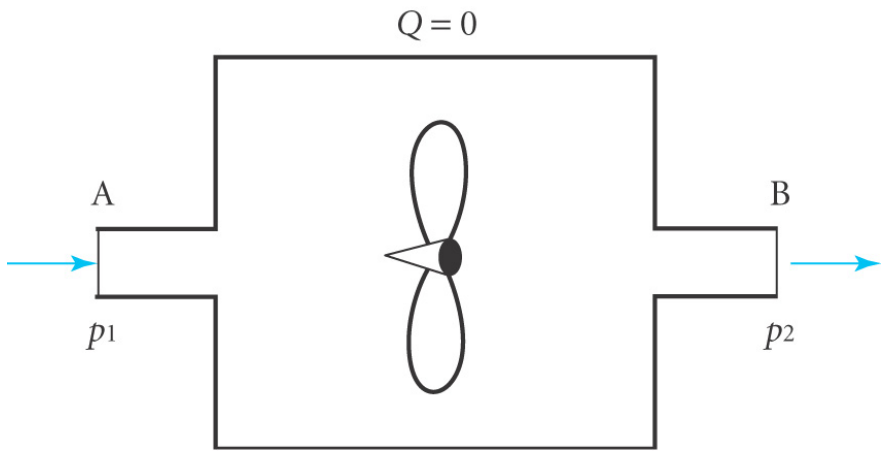


Figure 8: Schematic representation of an ideal adiabatic compressor or turbine.

Again it also follows that if a unit mass of ideal gas is heated from T_1 to T_2 at *constant pressure* then the heat into the system is given by $Q = c_p(T_2 - T_1)$. Finally, it is usual to define γ as the ratio of the specific heats, i.e. $\gamma = C_p/C_v = c_p/c_v$.

There are two further thermodynamic state functions in addition to U which are useful in describing the thermodynamics of thermal systems particularly those that cannot be described by the ideal gas law such as real gases and fluids (e.g. the water/steam system), namely the enthalpy H and the entropy S. (Again lower case symbols are used for U, H, S and V if they refer to unit mass of the thermal system. The notation is not always entirely consistent).

The **enthalpy** H (and the (specific) enthalpy (h)), which is extensively tabulated, are defined for gases by :

$$H = U + pV \text{ and } h = u + pv \text{ such that } dh = du + pdv + vdp \ .$$

Enthalpy is useful when heat transfer occurs at *constant pressure*, then :

$$dQ = dU + pdV \text{ integrates to } Q = U + pV = H \text{ or } Q = u + pv = h \ ,$$

so that at *constant pressure* the heat into a system is given by $Q = H_2 - H_1$.

Enthalpy is also useful in adiabatic processes. Consider an adiabatic compressor or turbine, shown schematically in Figure 8. If a unit mass of fluid of volume v_1, pressure p_1 and internal energy u_1 moves into a vessel containing a turbine then the fluid energy entering the vessel is $u_1 + p_1v_1$. Similarly if a unit mass of fluid of volume v_2, pressure p_2 and internal energy u_2 leaves the vessel then the fluid energy leaving is $u_2 + p_2v_2$. If conditions are *adiabatic* , i.e. no heat enters or leaves the system, then the work done by unit mass of the system on the turbine W_t is given by :

$$W_t = u_1 + p_1v_1 - (u_2 + p_2v_2) = h_1 - h_2 \ .$$

Entropy is a function of state that applies to irreversible as well as reversible thermodynamic processes. In an irreversible process the initial and final states can still be described by well defined state variables such as p, V, T etc. Entropy is again extensively tabulated. The entropy S is defined by $\int \frac{dQ_R}{T}$ along a *reversible* path that connects the initial and final states as defined by the state variables where T is the absolute temperature. For a reversible path $dS = dQ/T$ and $dQ = TdS$ so that the second law of thermodynamics reads :

$$TdS = dU + pdV \ .$$

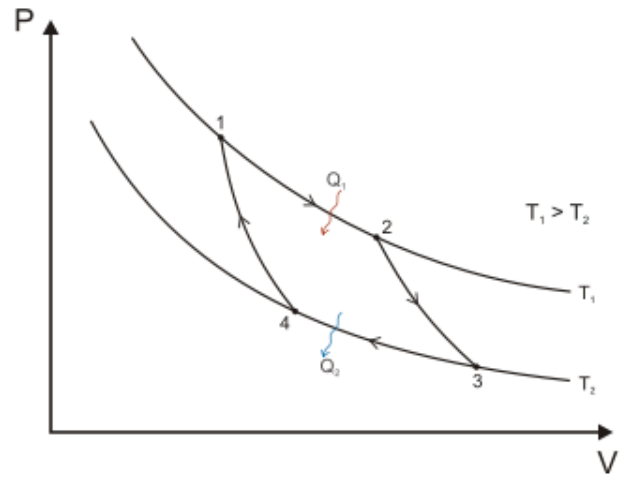


Figure 9: The Carnot Cycle on a pV plot.

Now the above relation is between state functions S, U and state variables T, p and V so provided the initial and final states are well defined by the state variables the above is valid for *irreversible* processes occuring between these states as well. This formulation of the second law of thermodynamics is one of the triumphs of classical 19th. century physics.

In order to convert heat into useful work a thermodynamic heat engine must be operated in a closed cycle so that the working gas (or fluid) returns repeatedly to its initial thermodynamic state. As a prototype of a heat engine we consider the Carnot cycle where an ideal gas contained in a cylinder, closed at one end and a movable, friction free piston is at the other end. At a constant temperature T_1 (or T_H) heat Q_1 (or Q_{in}) flows (isothermally) into the cylinder causing work to be done on the piston, the internal energy $U(T_1)$ remaining the same as the temperature remains constant. The gas is then allowed to expand adiabatically (i.e. no heat enters or leaves) such that more work is done on the piston, the temperature falling to T_2 (or T_C). The gas is compressed isothermally, the piston doing work on the gas and $Q_{out} = -Q_2$ of heat flows out of the system. Finally an adiabatic compression returns the gas to its initial state. The cycle is shown on a pV plot in Figure 9 and on a TS plot in Figure 10 (note that there is a correpondence between 1234 and ABCD in these two plots).

For any reversible cycle we can write $Q_{in} - Q_{out} = W_{cycle}$ since $\Delta U_{cycle} = 0$ because at the end of the cycle we return to the same initial condition where the internal energy U is the same. Thus the efficiency of a reversible heat engine (i.e. the work out for the heat put in), ϵ may be defined as :

$$\epsilon = \frac{W}{Q_{in}} = \frac{Q_{in} - Q_{out}}{Q_{in}} = 1 - \frac{Q_{out}}{Q_{in}} \ .$$

Heat engines can also be run in reverse so that the work input extracts heat at a lower temperature T_C and rejects heat at a higher temperature T_H such as in a refrigerator. In a heat pump, heat is 'pumped' from a lower temperature to heat at a higher temperature. The coefficient of performance COP is the ratio of the heat at the higher temperature divided by the work put in, i.e. :

$$\mathrm{COP} = \frac{Q_H}{Q_H - Q_C} \ .$$

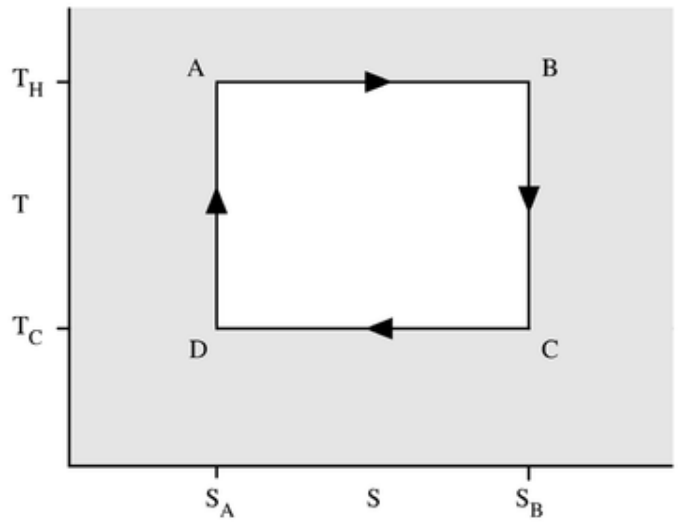


Figure 10: The Carnot Cycle on a TS plot.

Since we know that for a perfect gas $pV = nRT$ and $pV^\gamma = \text{const}$ for adiabatic changes, Q_{out} and Q_{in} may be calculated and $Q_{out}/Q_{in} = T_C/T_H$ for the Carnot cycle which yields the simple expressions :

$$\epsilon_{\text{Carnot}} = 1 - \frac{T_C}{T_H} \text{ and } \text{COP}_{\text{Carnot}} = \frac{T_H}{T_H - T_C} \ .$$

which should be remembered or at least quickly recalled using common sense. These results can also be shown to follow directly from the TS plot shown in Figure 10.

The Carnot engine has the maximum efficiency possible and indeed any cycle that is truly reversible has the same efficiency as the Carnot cycle. But truly reversible cycles cannot be achieved because it demands that every path in the cycle be reversible, and for gases the Carnot cycle cannot be reproduced easily in practice. Practical cycles are the Stirling cycle which powers pumps in the developing world, refrigeration equipment in space crafts and turbines in a large Spanish solar power plant, the Otto and Diesel cycles which are the basic cycles of internal combustion engines and the Brayton (or Joule) cycle which is the basic cycle for the turbo jet engine and for gas turbines used in electricity generation.

5.2 The Brayton Cycle and Gas Turbines

It is clear from the Carnot efficiency formula that it is an advantage to make the temperature at which heat flows into the cycle to be as high as possible. As we shall see later there are physical and practical limitations as to how high this temperature can be in water/steam power plants. In a gas turbine the gaseous products of combustion are typically 1300-1600 ° C and the metal turbine blades are covered with ceramic and the assembly is water cooled so that the temperature is kept below the metallurgical limit. Gas turbines for electricity generation were developed from jet engines.

Gas turbines operate in the Brayton cycle as shown in Figure 11. Air enters the compressor (a) at atmospheric pressure and is compressed adiabatically to around to around 10 - 20 bar (b). It is then mixed with fuel in the combustion chamber at constant pressure producing hot combustion gases that do work adiabatically on the turbine (c). The exhaust gases are then vented into the atmosphere unless the exhaust is to be used in a ‘combined cycle gas turbine’ (CCGT - see later). The efficiency of the Brayton cycle

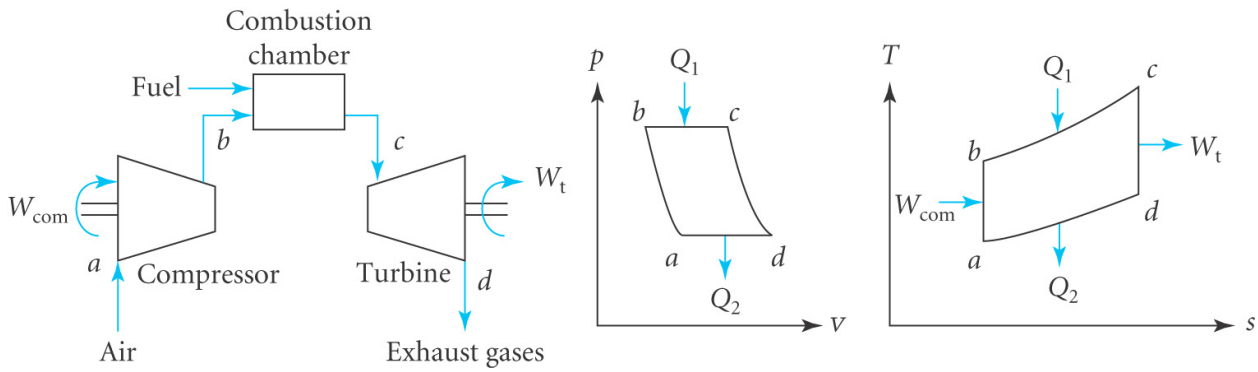


Figure 11: The Brayton cycle.

assuming the ideal gas law and that all processes are reversible can be easily calculated and is given by :

$$\epsilon_{brayton} = 1 - r^{-\left(\frac{\gamma-1}{\gamma}\right)} \text{ where } r = \frac{p_b}{p_a} = \frac{p_c}{p_d} \ .$$

The Brayton cycle is not truly reversible since heat is added (and rejected) when the temperature is not held constant. The Carnot efficiency calculated using the highest temperature at the input to the turbine and the lowest at the entrance to the compressor will be greater than that calculated from the ideal Brayton efficiency calculated from the pressure ratio. (In any case the Brayton cycle is not strictly comparable with the Carnot cycle.) The turbine runs at $3000(3600$ in the US) revolutions per minute so as to match the frequency of the AC supply. In practice efficiencies approach 0.4 since there are losses in the turbine and the ideal Brayton cycle cannot be achieved. A Gas turbine has the advantage of being readily switched on and off so can be used to respond to sudden demand fluctuations. As we shall see shortly the CCGT cycle in which the gas turbine is combined with a steam turbine is considerably more efficient for electric power generation.

We end this section with a diagram illustrating a turbofan jet engine (Figure 12. The compressors and turbines all rotate on a common axis but not all of the air is combusted. Some of the air is 'by-passed' and the fan attached also to the common shaft accelerates this air so at the exhaust of the engine lower density hot gases and cooler but more dense air provide the thrust for the aircraft.

5.3 Closed Cycle for a Steam Power Plant

Since water is cheap, fluid, and has relatively high density (1000 kg m^{-3}), specific heat (4.18 kJ kg^{-1} K^{-1}) and latent heat of vaporization (2260 kJ kg^{-1} at $100°$C) it has been the obvious choice to transfer heat and to be the working fluid in steam plants since the beginning of heat technology. However explosive conditions can result if water comes into contact with hot metals. The resultant phase change to steam whose volume is more than a thousand times greater than that of water can be explosive and hot steam can react with zirconium and graphite to produce hydrogen. Water's critical point (see below) also imposes conditions on its use in nuclear reactors.

In order to convert useful work in a steam power plant, the working fluid undergoes a *change of phase* at different stages in a closed cycle (ie the working fluid is reused), from liquid water, to a two-phase mixture of water and steam, to dry steam and back to liquid water. The key stages in the cycle are (see Figure 13) :

- in the *compressor*, work W_{com} is done on the system to compress cold water from sub atmospheric pressure to high pressure;
- in the *boiler*, heat Q_1 is added to the system to convert cold water to steam;
- in the *turbine*, work is done adiabatically by the steam on the turbine blades;

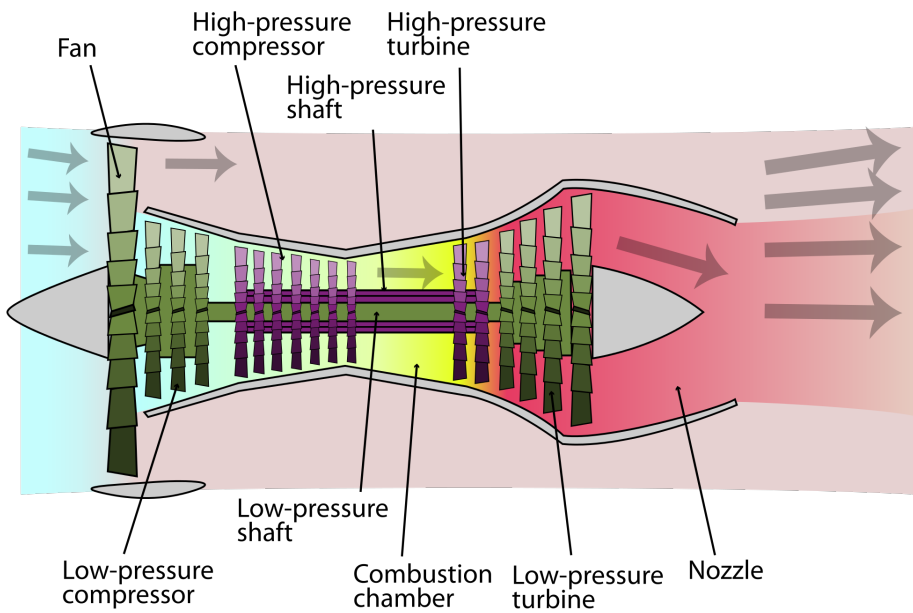


Figure 12: Turbofan jet engine.

- in the *condenser*, heat Q_2 is lost from the system to the environment in converting steam back into cold water.

Practical cycles such as the Rankine cycle (see the next section) will have variations on the above. We must now discuss the thermal properties of water and steam which are described in the $T-s$ diagram (Figure 14). There are three distinct regions of interest in this plot :

I. Water (note that the water region is confined to be quite close to the I/II boundary).

II. Two-phase mixture of water and steam.

III. Dry steam.

The bell shaped curve represents the phase boundary which has to be determined experimentally. Let us follow the isobar ABCD which is at atmospheric pressure. Along AB the water is being heated and gains entropy and enthalpy. At B the water begins to boil at the temperature 100°C and along BC the fluid absorbs latent heat at 100°C until at C all the water is converted to steam. At any point along this isobar there is a mixture of water and steam characterised by the 'steam quality' ratio which is the ratio of the mass of steam to the total mass of water plus steam. The line CD represents superheated steam. Assuming that water is incompressible and that its specific heat is temperature independent, and that steam is an ideal gas then it is a simple matter to evaluate s and h for fluid and steam with respect to their values at 0°C along the isobar ABC and D from the expressions for ds and dh by integration.

It is instructive to calculate the enthalpy and entropy changes along the path ABCD starting from A at 0°C or 273 K. For the enthalpy change Δh_{AB} we have for an incompressible liquid ($\Delta(pV) = 0$) and :

$$\Delta h_{AB} = \Delta u_{AB} = c_V \Delta T_{AB} = c_V (T_B - T_A) = 4.19 \times 100 = 419 \text{ kJ kg}^{-1} \ .$$

For the entropy change Δs_{AB} we have :

$$\Delta s_{AB} = \int_A^B \frac{dQ}{T} = \int_A^B \frac{d(u+pv)}{T} = \int_A^B \frac{du}{T} = \int_A^B \frac{c_V dT}{T} = c_V \ln \frac{T_A}{T_B} = 4.19 \ \ln \frac{373}{273} = 1.308 \text{ kJ kg}^{-1} \text{ K}^{-1} \ .$$

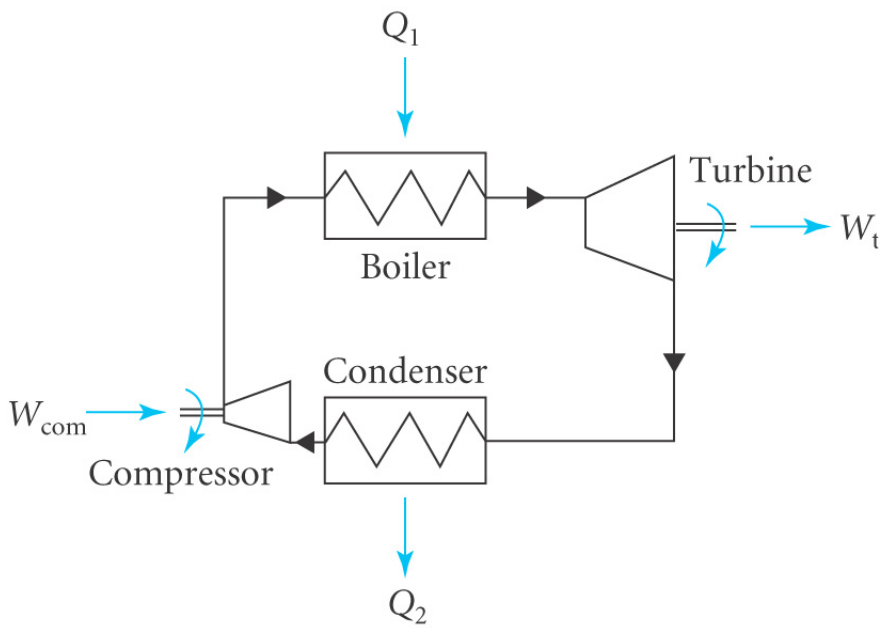


Figure 13: Plant lay-out of a thermal power station.

(You might notice that the expression for entropy contains the term $\ln T_B$ but as absolute zero is reached the specific heat becomes linear in the absolute temperature T, removing the logarithmic singularity and enabling absolute values of the entropy to be calculated.)

Along BC water is converted from the liquid to the gaseous state at 100°C by the input of 2260 kJ kg^{-1} of latent heat (L) so that :

$$\Delta h_{BC} = 2257 \text{ kJ kg}^{-1} \text{ and } \Delta \mathrm{s_{BC}} = \frac{\mathrm{Q}}{\mathrm{T_{BC}}} = \frac{2257}{373} = 6.05 \text{ kJ kg}^{-1} \text{ K}^{-1} .$$

Now consider the isobar for $p > p_{\mathrm{atmos}}$. The water starts to boil at a higher temperature and the required latent heat differs from the value at 100°C - both have to be measured.

Beyond a pressure of 218 bar and temperature of 374°C known as the *critical point* it is not possible for the liquid phase to exist. In water cooled nuclear reactors it is vital that there is water as in the Pressurized Water Reactor (PWR) or a two phase mix of water and steam as in the Boiling Water Reactor (BWR) in the reactor core (see later) so it is essential to remain below the critical temperature. In the BWR the maximum coolant temperature and pressure are 285°C and 75 bar and 315°C and 155 bar in the PWR which clearly limits the thermodynamic efficiency.

Provided the entropy and enthalpy are known in every region of the phase diagram then the 'steam quality' and the efficiency of any cycle can be determined. However its not quite as simple to determine s and h as indicated above along the path ABCD because the temperature and latent heats for phase changes need to be measured and water is not entirely incompressible, specific and latent heats are not constant throughout the range of T and p and steam does not behave like an ideal gas particularly near the crtical point. So extensive 'steam look up tables' have been developed to describe the real water/steam system.

Figure 15 shows a steam power plant operating in an idealised Carnot cycle which is the most efficient cycle possible. The boiler is at $T_1 = 352$°C and $p = 170$ bar and the condenser is at $T_1 = 30$°C and $p = 0.04$ bar. Under isothermal conditions (latent) heat flows into the water/steam mixture at temperature $\mathrm{T_1}$ along ab, the steam at b is allowed to expand adiabatically doing work on the turbine partially condensing on the way (bc), the resultant mixture is incompletely condensed in the condenser at T_2 (cd) and then is compressed in the compressor (da). From the temperatures the Carnot efficiency is 0.52 and the same result is obtained

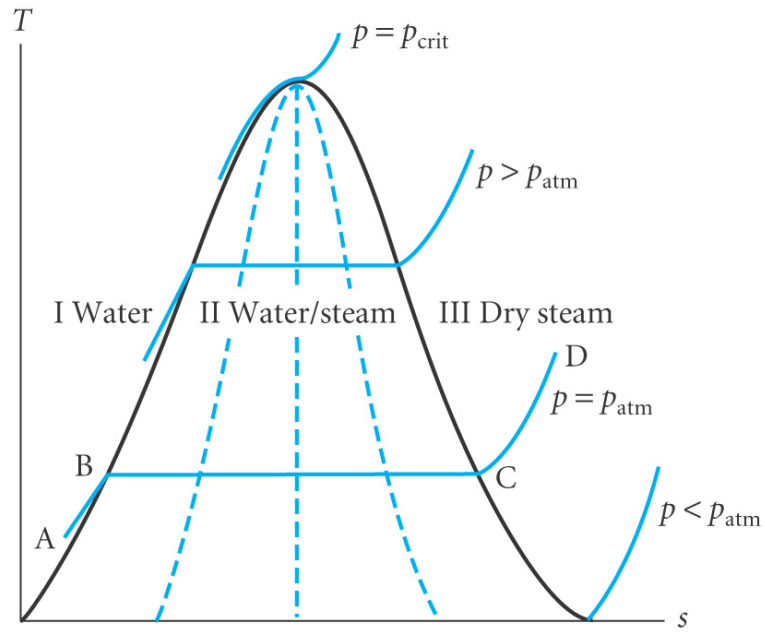


Figure 14: T - s diagram for water and steam (not to scale).

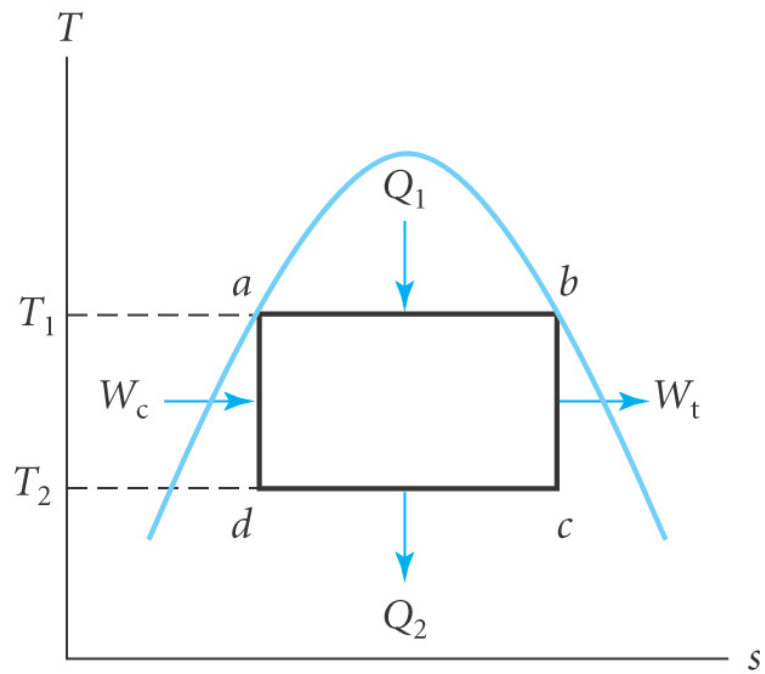


Figure 15: T - s diagram for steam powerplant operating in a Carnot cycle .

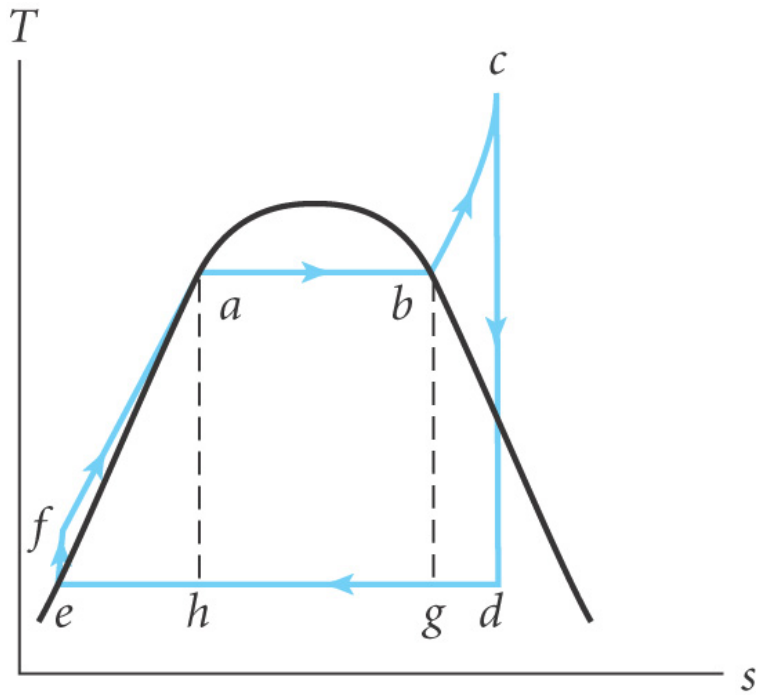


Figure 16: The Rankine cycle without reheat

by the enthalpy and entropy calculations made in [2]. Even if this idealised cycle could be realized in practice, it is impractical for two reasons. Firstly the Carnot cycle requires the temperature of the boiler to remain constant which restricts the operation of the cycle to be within the two-phase mixture of water and steam. The maximum temperature is thus restricted to the maximum temperature of the two phase boundary. Secondly water droplets from the condensing water in the adiabatic expansion phase passing through the turbine would damage the turbine blades.

5.4 The Rankine Cycle

The shortcomings of the Carnot cycle is overcome in the Rankine cycle (Figure 16) named after the famous Scottish thermodynamicist Thomas Rankine. In the Rankine cycle without reheat all the steam in this cycle is converted into water in the condenser (de) before entering the compressor. The compressor increases the pressure of the water adiabatically (ef) before the water enters the boiler. In modern steam plant boilers are constucted in three distinct sections, each made with a different grade of steel, using cheaper steels in the lower temperature sections and more expensive steels in the higher termperature sections. Indeed to avoid excessive metallurgical damage the highest temperature must be limited to $650°$C. In the *economizer* section (fa) water is heated at high pressure until it starts to boil. In the *evaporator* section (ab) the two phase mixture is heated at constant pressure until all the water has been converted into dry steam. The dry steam is then heated at constant pressure in the *superheater* section of the boiler (bc). The dry, superheated steam enters the turbine at high pressure and does work adiabatically on the turbine blades (cd). On leaving the turbine the wet steam enters the condenser (at subatmospheric pressure), where it condenses on the cold outer surfaces of a large bank of condenser tubes containing cold water from the external environment. In a modern power station the turbine is divided into high pressure, intermediate pressure and low pressure sections and the steam reheated between each section to avoid the production of water droplets - this is the so called ‘Rankine cycle with reheat’.

For a Rankine cycle working between $600°$C and $30°$C the Carnot efficiency would be 0.52 and the

Rankine efficiency is worked out from steam tables in example 2.6 in [2] to be 0.45. In practice of course the efficiencies are lower than this - fossil fuel power stations average 33% thermodynamic efficiency and thus produce a large amount of low grade heat. Flue Gas Desulpherization (FGD) is fitted as standard to most coal fired stations and accounts for up to 10% of the lost efficiency.

State of the art coal fired power stations, the so called 'Advanced Super Critical' (ASC) coal power plants run at 620°C, well above the critical point of water and require specialist steels for boiler construction. If heat is rejected at 30°C the Carnot efficiency would be 0.66 and in practice efficiencies of $0.46-0.48$ are achieved. These efficiencies will be considerably lower when Carbon Capture and Sequestration (CCS - see later) technology is fitted to the exhaust stream of the power station. Steam turbines are manufactured to generate up to $2000\ \mathrm{MW_e}$ of electricity. In future they are likely to be used in conjunction with water cooled nuclear reactors where the steam temperatures are not particularly high (see above), coal fired stations hopefully with CCS or combined with high temperature gas turbines in the CCGT cycle with CCS technology which are described in the next two sections. Steam boiler operations cannot respond to sudden changes to power demand but can respond to forseen diurnal demand such as that associated with the morning rush hour.

5.5 Combined Cycle Gas Turbine (CCGT)

The overall efficiency of a gas turbine operating on the Brayton cycle can be increasesd by feeding the exhaust gases into a steam power plant operating on the Rankine cycle (see Figure 17). A CCGT operating between 1300°K and 300°K would have an ideal Carnot efficiency of 0.76 - in practice efficiencies of up to 0.60 are achieved. Typical CCGT plant combine a $400\ \mathrm{MW_e}$ gas turbine with a $200\ \mathrm{MW_e}$ steam turbine.

Efficiencies could be higher still if the 'waste heat' at around 30°C could be utilised in domestic heating or in industrial processes such as paper making. Such schemes known as Combined Heat and Power (CHP) do exist particularly in Russia and Finland but in the UK domestic heating would require extensive piping and it would be difficult to match the cooling demands of the power generation to the seasonal demand for domestic heating.

6 Carbon Capture and Sequestration (CCS)

It is clear that the burning of fossil fuels cannot be regarded even as an interim solution to climate change unless the CO_2 inevitably generated is one way or another removed from the exhaust gases. CCS technology has yet to be demonstrated on the scale of the emissions of a large power plant. In the UK a pilot plant to demonstrate carbon capture and sequestration at the $2400\ \mathrm{MW_e}$ Longannet coal fired power station in Fife has been abandoned by the industrial partners because government funds were insufficient to complete the project. There were no takers either for EU funding to build such CCS plants in Europe. The $110\mathrm{MW_e}$ Boundary Dam Power Plant in Canada is the first commercial station with CCS in operation. Some of the CO_2 is being used to push more oil out of depleted reservoirs - hardly a contribution to the reduction of global CO_2! The rest of the CO_2 is being geologically sequestered.

In CCS plants, the flue gases are bubbled through the solvent monoethanolamine (MEA) in a packed absorber column. Afterwards the solvent passes through a regenerator unit where the absorbed CO_2 is stripped from the solvent by counterflowing steam at $100-120°$ C. Water vapour is condensed, leaving a highly concentrated CO_2 stream which may be compressed and pumped for sequestration in depleted North Sea oil reservoirs. Depleted North Sea oil reservoirs in Europe and saline acquifers in the US could store 100 years of CO_2 emissions from both continents provided CCS can be made to work commercially.

In order to reduce the volume of the flue gases so that the size, and thus the capital and running cost of the absorber column is reduced it has been suggested that the combustion chamber be fed oxygen rather than air. Unfortunately there is an immediate 15% penalty in separating the oxygen from the air. Another approach being investigated in the US is the process by which 'syngas' consisting of hydrogen and carbon

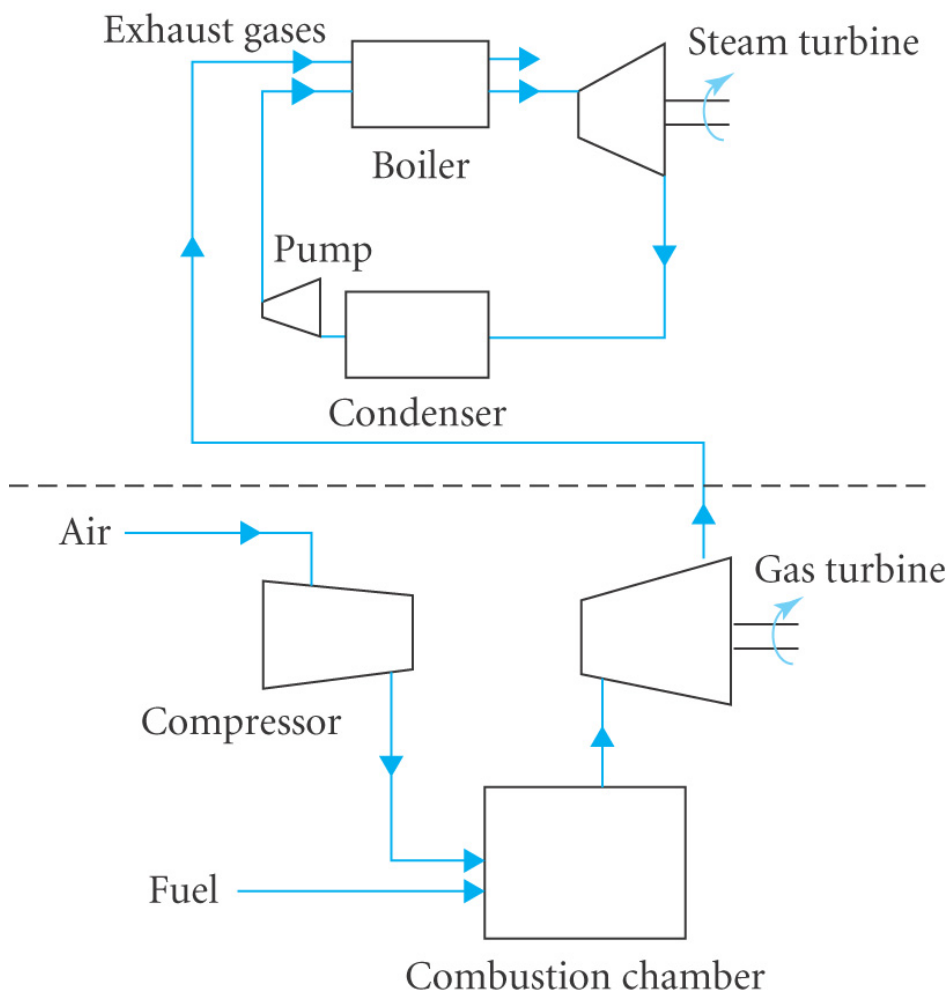


Figure 17: The CCGT cycle.

monoxide is obtained from the gasification of coal, is fed into the CCGT cycle. This is known as the Integrated Gasification Combined Cycle (IGCC) and has high capital costs.

Another way to generate electricity in a carbon neutral manner is to burn wood or other biomass. Although at least two such plants (see below for developments at the DRAX power station) are being constructed in the UK, it is unclear how big an impact biomass can have on the UK's energy generation infrastructure.

7 Fossil Fuels : Conclusions

Based on statistics produced by BP in 2007, at current consumption rates gas reserves would have been depleted in about 65 years whilst coal was good for another 300 years. However the exploitation of shale gas by using (a lot of) water to fracture deep rocks will make the US not dependent on imported gas and global prospecting will significantly extend the depletion date. Indeed the fall in US coal prices has resulted in the UK coal burn to increase, replacing Liquid Natural Gas (LNG) supplies made more expensive following the Fukushima nuclear accident. Certainly fracking has changed the global politico - economic balance and in the short term will lead to a reduction in CO_2 emission by displacing the coal burn. Fracking remains controversial particularly in Europe (in the US Dick Cheney exempted Oil and Gas companies from environmental legislation) and there are claims that the process releases some methane, which is a much worse greenhouse gas than CO_2, into the atmosphere. Large amounts of natural gas (more than current reserves) are locked up in clathrate deposits in deep ocean or lakes and in permafrost regions. Some exploratory prospecting is in progress because it is imperative that methane gas should not escape since it a far worse greenhouse gas than CO_2. (The ultimate catastrophe in Global Warming would be a temperature rise so great that the clathrates release methane into the atmosphere. There is already concern on this score given the retreat of the arctic ice sheet in recent years.)

In the UK Centrica has made a long term contract to import LNG from fracked gas in the USA at a cost of $2\mathrm{p\ per\ kWh_t}$ to which must be added the transport costs to bring the LNG to the UK by specialized double hulled refrigerated ships. Fracking techniques promise also to extract oil from underground shale oil deposits - there is already large scale extraction by this process in the state of Dakota.

Major oil companies are also involved in the strip mining of the Athabasca oil sands in Alberta, Canada, yielding bitumen which can be upgraded to synthetic crude oil (SCO). This oil reserve is second only to that of Saudi Arabia. After stripping the trees and brush and the top layer of sand, clay and musket (peat bog), 2 tonnes of shale sand are needed to be removed to produce 1 barrel (0.136 tonne) of SCO as the end product. The process requires 2 to 4.5 times the volume of water to that of the oil, and on a 'well to wheel' basis produces 1.3 to 1.7 times the CO_2 emission as oil conventionally produced from wells. The water ends up in tailing ponds and processes are being developed so that the water may be recovered and re-used and the tailing ponds reclaimed to allow land reclamation to follow continuously 7 to 10 years after the mining operation. Production cost of a barrel of SCO is $\$27$ and is still very profitable to the oil companies. Since the Canadians do not wish to be a part of a Carbon Tax mechanism which would make the process uneconomic they have withdrawn from the Kyoto protocol.

Over and above issues of the sustainability of the fuel resources is the question whether CCS technology can be made to work on a large scale (see above). But new techniques are being tested which promise to release the energy potential of inaccessible coal deposits and at the same time capture a substantial amount of the CO_2 released. One such project is attempting to consume inaccessible coal reserves under the Swansea bay by drilling vertically and then horizontally into the coal seam, injecting oxygen and water and causing the coal to combust. The gas produced, known as syngas and carrying approximately half the energy content of natural gas (methane CH_4), is brought to the surface through another drill hole and consumed in a gas turbine (or CCGT) to produce electricity. It is anticipated that the CO_2 produced could be captured by CCS

technology and then stored in the depleted coal seam and perhaps in surrounding porous rock formations.

In the UK older coal fired power stations are being retired because they do not meet EU standards for sulphur, SO_x and NO_x emissions. At DRAX half of the six coal fired boilers which feed six $660\mathrm{MW_e}$ turbines are being converted to wood pellet burning. This is part of the UK effort to meet the EU target of 20% renewable electricity generation by 2020. When on stream the electricity generated will attract 1 Renewable Obligation Certificate (ROC) which guarantees the company the floor price of $£42$ per $\mathrm{MW_e}$ (or 4.2 p per $\mathrm{kW_e}$) generated, which is currently below the wholesale cost to the customer.

However recent easing in the supply position of natural gas and crude oil through fracking and the exploitation of shales should not blind us to the absolute necessity of finding alternatives to fossil fuels. At best fracking should be regarded as a bridge to a largely fossil free energy supply future. Even if CCS technology should be made to work this would not apply to the transport sector which currently is a major user of oil in particular.

The following table taken from recent DECC reports summarises the relative costs of fossil fuel technology and should be studied and understood in terms of the preceding discussions. **Since April 2013 there is small carbon tax on current UK power supply which was expected to rise, but there is talk of its suspension to meet the political imperative 'to reduce the cost of living'. The EU carbon price is extremely low which indicates the failure of the carbon pricing market**. 'Cap. Cost' is the capital cost to *install* $1\ \mathrm{kW_e}$ of power generating capacity. 'Am. Cap.Cost' is the ammortised capital cost over the forseen plant lifetime, (and all other costs), are per $1\ \mathrm{MWh_e}$ energy *generated* and are easily converted to costs in $\mathrm{p/kWh_e}$ by dividing by ten. These costs add up to the cost per $\mathrm{kWh_e}$ generated, expressed in $\mathrm{p/kWh_e}$ at the bottom of the table. Note that ASCOCCS refers to the Advanced Super Critical coal plant fitted with CCS and fired with oxygen.

	CCGT	CCGT CCS	ASCOCCS
Cap. Cost ($£/\mathrm{kW_e}$)	808	1370	3000
Am. Cap. Cost ($£/\mathrm{MWh_e}$)	9	21	37
Fuel ($£/\mathrm{MWh_e}$)	49	56	36
Carbon ($£/\mathrm{MWh_e}$)	18	4	5
CO2 Capture/Storage ($£/\mathrm{MWh_e}$)	0	4	5
Decommision/ Waste	0	0	0
O+M	4	10	22
Elec. Gen. Cost ($\mathrm{p/kWh_e}$)	8	9.5	10.5

Table 2: Summary of fossil fuel power plant costs.

8 Alternative Methods of Energy Conversion

In the above section we have considered how thermal energy can be converted to useful work using cyclical heat engines whose ideal efficiencies cannot exceed the Carnot efficiency. The fundamental limitation of even ideal heat engines arises because heat has to be rejected at the lower temperature so as to complete the cycle. However heat energy can be converted to useful work (and vice versa) in a non cyclical manner thus avoiding this fundamental limitation. The starting point is the maximum amount of work done by a system which is in contact with a large heat reservoir held at temperature T_0 through a moveable diathermal wall. At the beginning and at the end of any process taking place in the **system** we consider that the temperature and pressure are T_0 and p_0 - this would be the situation of changes taking place at room temperature and atmospheric pressure. There is no limitation on the system during the process.

For the combination of heat reservoir and system the entropy change ΔS has to be greater than, or for a

reversible pocess, equal to zero. Thus :

$$\Delta S = \Delta S_{sys} + \Delta S_{res} \geq 0 \ .$$

Now imagine a quantity of heat Q leaving the reservoir and going into the system; then :

$$\Delta S_{sys} - \frac{Q}{T_0} \geq 0 \quad \text{or} \quad Q - T_0 \Delta S_{sys} \leq 0 \ .$$

For the system we have $Q_{sys} = \Delta U_{sys} + W_{sys}$, therefore $W_{sys} \leq -(\Delta U_{sys} - T_0 \Delta S_{sys})$.

From now on we drop the subscript sys but recall that all state variables and state functions on the left of the inequality are those of the system, and the temperature and pressure at the beginning and end of the process are the same at T_0 and p_0. Now the work done by the system, W, is not all useful work since for example gases might be produced which would exert a pressure on the diathermal wall and work has to be done in the expansion of the system volume. Thus we take the useful work done *by the system, under the above conditions* to be given by $W_{use} = \Delta W - p\Delta V$. Then the useful work can be given by :

$$W_{use} \leq -(\Delta(U + pV) - T\Delta S) = -(\Delta H - T\Delta S) = -\Delta G \text{ where } G = U + pV - TS \ ,$$

where G is known as the Gibbs Free Energy and G is expressed in terms of H and S in the second expression above since extensive tabulations exist for H and S.

8.1 Electrolysis, Fuel Cells and Batteries

Parts of this section are highly relevant to the transport sector. Recall that in 2007 the UK primary power consumption of $125 \text{ kWh d}^{-1}\text{p}^{-1}$ (or 5 kW p^{-1}), $40 \text{ kWh d}^{-1}\text{p}^{-1}$ is the power devoted to transportation whose source is largely fossil fuel in situations difficult to envisage carbon capture.

8.1.1 Electrolysis

If a charged battery is connected across the terminals of a cell containing water plus some sulphuric acid for electrical conduction then hydrogen gas will appear at one of the terminals and oxygen at the other, the underlying reaction being the reverse of $\text{H}_2 + 0.5\text{O}_2 \rightarrow \text{H}_2\text{O} + 143 \text{ MJ}$ per kg. Hydrogen (39.7 kWh kg^{-1} or 286 kJ mole^{-1}). Electrolysis occurs in other liquids such as Methanol and Ammonia and in molten salts where the process is important industrially.

We can use the expression for the Gibbs free energy given above, or we can arrive at the same result through a more directly physical argument, as follows. This electrolysis process requires 286 kJ mole^{-1} energy to be supplied to break up the water molecule, and to do the work done against the atmosphere by the produced gases. But in creating the gases from water the entropy increases and the product $T_0\Delta S$ is 49 kJ at 298°C (you have to get this from tables of standard entropies). Now this energy comes from the surrounding heat reservoir and the required work done by the battery is $286 - 48 \text{ kJ mole}^{-1}$, i.e. 237 kJ mole^{-1}. The -48 kJ mole^{-1} term is the $-T\Delta S$ term in the expression for the Gibbs free energy.

In one mole of Hydrogen there are 6×10^{23} molecules and the charge transferred by each electron is 1.6×10^{-19}C(oulombs) so that when a mole is electrolysed a charge of $\text{z} \times 9.65 \times 10^4 \text{C mole}^{-1}$ where z is the number of valency electrons, 2, in this case is transferred through the electrolytic cell by the EMF of the battery. Thus $\text{EMF} \times 2 \times 9.65 \times 10^4 \text{ J} = 237 \times 10^3 \text{ J}$ showing that the minimum EMF is 1.22 V.

This sounds too good to be true - the application of 1.2 V to the electrode and 237 kJ mole^{-1} of electrical energy produces a mole of hydrogen and half a mole of oxygen containing 286 kJ mole^{-1} of energy, the balance coming from the surroundings. But this ideal conversion efficiency is reduced to about 60% in real life by the internal resistance of the cell and other subtle effects.

8.1.2 Fuel Cells

Fuel cells are cells which work in reverse to electrolysis - gases hydrogen and oxygen (normally directly from the air) are pumped into the electrodes between which an EMF appears, driving a current through the cell and an external circuit. Electrolytes can be liquid or solid and are the subject of current research, since the promise of converting gaseous fuel to electricity, thereby defeating the Carnot limit is extremely attractive, particularly as the hydrogen oxygen cell produces only water. The first fuel cell was developed by the Welshman William Grove in 1839. The 1.5 moles of the incoming gases liberate $286\ \mathrm{kJ\ mole^{-1}}$ of energy when they convert to water in the cell - this is the input energy into the cell. In this change the entropy of the gases decrease and the $\mathrm{T\Delta S} = -49\ \mathrm{kJ\ mole^{-1}}$ term has to be subtracted from the incoming energy (or enthalpy) to yield the $237\ \mathrm{kJ\ mole^{-1}}$ of useful work, i.e. the electrical work done. So repeating the argument in the previous paragraph on electrolysis, a voltage of 1.22 V will appear across the cell terminals. The theoretical efficiency of this conversion to work is $237\ \mathrm{kJ\ mole^{-1}}/\ 286\ \mathrm{kJ\ mole^{-1}} = 0.83$. In practice of course the efficiencies are lower than this, typically 50% when producing 0.8 V at a current of 1 Amp.

Hydrogen fuelled cells produce carbon free electricity. Such cells nowadays consist of solid polymer membrane replacing a liquid electrolyte, which makes the costs very high. Around the world there are about 100 hydrogen fuel cell powered buses - several of them in London.

Solid Oxide Fuel Cells (SOFC) operate at $800 - 1000$°C and run on several gases including hydrogen and natural gas (methane). Such $10\ \mathrm{kW}$ units running on methane co-generate electricity and heat, but also $\mathrm{CO_2}$, and are used to power headquarter buildings, banks etc.

Electrolysis, hydrogen gas compression and fuel cells are certainly important components in the 'hydrogen economy' which only makes sense if the primary energy source is nuclear or renewable such as wind power since non electrolysis methods to make hydrogen produce $\mathrm{CO_2}$. At the end of the next section on Transport, a hydrogen fuel cell car is compared with a battery powered car, both of which are commercially available.

8.1.3 Batteries

Table 3 summarises the energy densities that can be stored by fuels and batteries *including the weight and the volume of storage containers* and it is apparent that for batteries these are much lower than those for petrol. Futhermore the cost of lithium-ion batteries for car transportation adds some $£11,000$ to the costs of the electric Ford Focus which has a range of about $110\ \mathrm{miles}$. Clearly developments in battery technology are required to lower costs and to improve the range of electric cars. Other issues are how often a Lithium Ion battery can be recharged and what can be recovered from spent batteries.

	kWh/kg	kWh/l
petrol	12	9
wood	5.4	2.7
coal(C)	6.5(9)	5.5
ethanol	7.5	6
methane	15.3	
hydrogen(H2)	1.9 (40)	1.9
hydrogen(34bar)	1.8	0.6
lead-acid	0.04	0.07
lithium-ion	0.135	0.3
Na-S	0.10	

Table 3: Calorific values (LHV) of Fossil Fuels in kWh per kg and KWh per litre, and Battery Energy Densities

Two of the most important criteria for batteries are :

- high energy stored per kilogram, and
- as large a number of charge/discharge cycles as possible.

The Tesla Corporatation who build battery driven electric cars in California are constructing a large plant near Reno, Nevada to produce $\mathrm{LiFePO_4}$ (Lithium Ferrous Phosphate) battery packs which store $70\ \mathrm{kWh_e}$, weighing 540 kg with an energy density of $0.130\ \mathrm{kWh_e\ kg^{-1}}$ at a cost of $\$220\ \mathrm{kWh_e^{-1}}$ and sufficient for a range of 240 miles. Tesla claim that this is the limit for battery power in view of the substantial weight of the battery, unless there can be improvements in the energy density.

The Oxis company in Culham near Oxford are claiming such an improvement. It claims that its Lithium Sulphur battery can store up to $0.500\ \mathrm{kWh_e\ kg^{-1}}$ the current aim being $0.350\ \mathrm{kWh_e\ kg^{-1}}$. Time will tell if this can be realised or not. The Alevo corporation is building a large plant in North Carolina to produce a 'container worth' of Lithium Phosphate battery to store $2\ \mathrm{MWh_e}$ and rated to produce power up to $1\ \mathrm{MW_e}$ and claim 40,000 charge/discharge cycles. Such units are designed to smooth the Grid power against supply/demand fluctuations. A 240 tonne Lithium Manganese Oxide battery, occupying 'three tennis courts' has been installed in Leighton Buzzard at a cost of $\pounds 520\ \mathrm{kWh_e^{-1}}$ which can store $10\ \mathrm{MWh_e}$ and produce power up to $6\ \mathrm{MW_e}$. Again this battery is for Grid smoothing.

Large Sodium Sulphur (Na-S) batteries working at 300°C have been installed in Japan and the US to store electricity at various points in the Grid so as to ensure a steady power output when supply and demand fluctuate. In West Virginia an 80 tonne battery storing $7.2\ \mathrm{MWh_e}$ of electrical energy and capable of discharging at a power of $1\ \mathrm{MW_e}$ has been operating for a number of years. The cost of this installation is around $\pounds 500$ per stored kWh or so it is not particularly cheap! The corresponding cost for a pumped storage scheme (see later notes) is around $\pounds 100$ per stored kWh but this cost does depend on operational details.

Hyundai is manufacturing 1000 hydrogen fuel cell cars in the first instance which have a power output up to $100\ \mathrm{kW}$ fuelled by two tanks each of weight $28\ \mathrm{kg}$ and each containing $1.456\ \mathrm{kg}$ of hydrogen at $700\ \mathrm{bar}$. The claimed range is 404 miles and the car costs $\approx \pounds 40,000$. The Hyuandai car can be compared with the Electric Ford Focus powered by a $170\ \mathrm{kg}$ Lithium Ion battery which has a claimed range of 100 miles and costs $\approx \pounds 33,000$.

Mackay has suggested using the batteries of the UK fleet of cars as backup to wind energy. It is is reasonable to suppose that fully charged batteries should store $40\ \mathrm{kWh}$ of electrical energy so that a fleet of 30 million cars would store $1200\ \mathrm{GWh}$, just enough for a five day lull in the wind (see later). There would be quite a few practicalities to sort out before implementation!

References

[1] "Sustainable Energy - without the hot air", David MacKay (2009).

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